

# Hydroponic Nutrients

Guide 4 · Nutrients, pH & Precision

*Masterblend · DIY · pH · Indoor*

*Version 2026.05 · Revised 2026-05-16*

# Contents

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| <b>About this guide</b>                                           | <b>5</b>  |
| <b>Part 1 — Masterblend</b>                                       | <b>8</b>  |
| 1.1 Why Masterblend 4-18-38 . . . . .                             | 8         |
| 1.2 Stock solution preparation . . . . .                          | 9         |
| 1.3 Mixing order — the strict A → Epsom → CaN-last rule . . . . . | 9         |
| 1.4 Per-gallon feeding charts by crop . . . . .                   | 10        |
| 1.5 Full nutrient profile at standard rate . . . . .              | 12        |
| 1.6 EC and pH daily protocol . . . . .                            | 12        |
| 1.7 Deficiency diagnosis table . . . . .                          | 13        |
| 1.8 Crop-specific optimisation . . . . .                          | 14        |
| <b>Part 2 — Locally-sourced DIY</b>                               | <b>17</b> |
| 2.1 Why this part exists . . . . .                                | 17        |
| 2.2 Target ppm profile for the three-part replacement . . . . .   | 17        |
| 2.3 Local sources . . . . .                                       | 18        |
| 2.4 Micronutrient sourcing . . . . .                              | 22        |
| 2.5 Feasibility scoring . . . . .                                 | 23        |
| 2.6 The 8-step DIY workflow . . . . .                             | 24        |
| <b>Part 3 — pH adjustment recipes</b>                             | <b>25</b> |
| 3.1 Why pH control matters . . . . .                              | 25        |
| 3.2 pH Down recipes . . . . .                                     | 26        |
| 3.3 pH Up recipes . . . . .                                       | 31        |
| 3.4 Comparison table . . . . .                                    | 34        |
| 3.5 Universal dosing protocol . . . . .                           | 36        |
| <b>Part 4 — Indoor environment &amp; precision</b>                | <b>38</b> |
| 4.1 LED basics . . . . .                                          | 38        |
| 4.2 VPD and CO <sub>2</sub> management . . . . .                  | 40        |
| 4.3 Sensor priority list . . . . .                                | 41        |
| 4.4 Automation . . . . .                                          | 42        |
| <b>Sources &amp; references</b>                                   | <b>45</b> |
| Local resources . . . . .                                         | 45        |
| Books & references . . . . .                                      | 45        |
| Online resources . . . . .                                        | 45        |
| <b>Closing</b>                                                    | <b>47</b> |



**RELATED GUIDES**

This guide is one of fourteen in the Dragoon Mountains Growing Guides Library: - **Guide 1 — Hydroponic Primer & Library Map** · the front door + master multi-crop calendar - **Guide 2 — Peppers in the Dragoons** · soil + hydroponic + variety-specific staking — the crop this nutrient program was tuned around - **Guide 3 — Bato Bucket Hydroponic Systems** · system build, water treatment, operations - **Guide 4 — Hydroponic Nutrients** (this guide) · Masterblend + DIY + pH + Indoor/LED - **Guide 5 — Grapes in the Dragoons** · Cochise County viticulture - **Guide 6 — Cool-Season Crops** · brassicas + alliums + greens + roots - **Guide 7 — Warm-Season Companions** · tomatoes, cucurbits, beans, corn, sweet potatoes - **Guide 8 — Herbs in the Dragoons** · annual + perennial · soil & hydroponic - **Guide 9 — Backyard Orchard in the Dragoons** · figs, pomegranates, jujubes, stone fruits - **Guide 10 — Pest, Disease & Wildlife** · IPM + Cochise wildlife exclusion - **Guide 11 — Foundations: Soil & Seed** · soil-building, composting, seed-starting, propagation - **Guide 12 — Harvest, Preservation & Storage** · canning, fermenting, drying, cellaring - **Guide 13 — Water in Cochise County** · sourcing, testing, treatment, rainwater, conservation - **Guide 14 — Season Extension Structures** · cold frames, low tunnels, hoop houses, high tunnels

## About this guide

*Four tracks · one nutrient profile · the same pepper at the end*

This is the longest reference in the Dragoon Mountains library, and it is deliberately structured as four parallel tracks that all converge on the same crop ppm targets. You can read it cover to cover, but most growers will live in one or two parts at a time.

| Part                                | Track                                           | When you read it                                                                     |
|-------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Part 1 — Masterblend</b>         | Purchased 3-part precision program              | Daily use. The canonical recipe and feeding charts.                                  |
| <b>Part 2 — Locally-sourced DIY</b> | Off-grid brews from Cochise County materials    | Project mode. Replacing one or all three Masterblend parts with foraged equivalents. |
| <b>Part 3 — pH adjustment</b>       | Acids and bases — purchased and homemade        | Every day, for as long as you run hydroponics.                                       |
| <b>Part 4 — Indoor environment</b>  | LED, VPD, CO <sub>2</sub> , sensors, automation | When you bring the system inside or under a roof.                                    |

**If you are new to hydroponic nutrients, start with Part 1 §1.1 (Why Masterblend 4-18-38) and work straight through Part 1 §1.6 (EC and pH daily protocol).** Do not jump to Part 2 (DIY foraging), Part 3 (acid/base recipes), or Part 4 (indoor LED precision) until the Masterblend program in Part 1 is working in your reservoir. Part 1 is the spine; the other three parts are wings off it. The 38 pages of this guide are arranged in priority order — read top-down on your first pass.

### Where do I start?

| If you...                                                                | Read this first                                                                                  |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Have a bag of Masterblend and want to mix your first reservoir           | <b>Part 1 §1.1 → §1.6</b> (Masterblend rationale, mixing order, per-crop charts, daily protocol) |
| Are seeing yellow leaves and need to diagnose                            | <b>Part 1 §1.7</b> (deficiency diagnosis table)                                                  |
| Have a reservoir already running but pH won't hold                       | <b>Part 3 §3.1 → §3.5</b> (acid / base chemistry and dosing)                                     |
| Want to brew your own potash, calcium, or magnesium from local materials | <b>Part 2</b> (DIY track) — but only after Part 1 is stable                                      |

| If you...                                         | Read this first                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------|
| Just brought the system into a garage or basement | <b>Part 4</b> (LED, VPD, CO <sub>2</sub> , automation)                                 |
| Are still choosing between hydroponic and soil    | Stop. Read <b>Guide 1</b> (the primer) and <b>Guide 3 §1</b> (why Bato buckets) first. |

Parts 1 and 2 target the **same nutrient ppm profile** — 116 N / 47 P / 188 K / 113 Ca / 41 Mg / 58 S, with the standard micronutrient package. (**ppm** is *parts per million* — milligrams of nutrient per liter of solution; the standard unit for reporting a hydroponic recipe.) Whether the potassium comes from a bag of *Masterblend 4-18-38* or from a bucket of leached oak ash, the crop does not know the difference. What changes is how much measurement, how much foraging, and how much consistency you sign up for.

### Abbreviations you will encounter

This guide leans heavily on a small vocabulary. Definitions appear inline at first detailed use, but the cheat sheet lives here:

- **EC** — *electrical conductivity*, in mS/cm (millisiemens per centimeter). A proxy for total dissolved nutrient concentration. Most fruiting crops run EC 2.0–2.8.
- **TDS** — *total dissolved solids*, in ppm. The same measurement as EC, different units; 1.0 mS/cm  $\approx$  500 ppm on the 0.5 conversion scale.
- **pH** — acidity / alkalinity, 0–14 scale. Target 5.8–6.2 for nearly every crop in this guide. Above 6.5, iron and manganese lock out.
- **PPFD** — *photosynthetic photon flux density*, in  $\mu\text{mol}/\text{m}^2/\text{s}$ . The instantaneous photosynthetic light hitting the leaf surface. (Part 4 §4.1.)
- **DLI** — *daily light integral*, in  $\text{mol}/\text{m}^2/\text{day}$ . PPFD totaled across the photoperiod — the total daily light dose.
- **VPD** — *vapor pressure deficit*, in kPa. The driving force for transpiration — how thirsty the air is. (Part 4 §4.2.)
- **DWC / NFT** — *deep water culture* (roots suspended in oxygenated reservoir) and *nutrient film technique* (thin stream of solution through troughs). The two non-bucket systems referenced in the per-crop tables.
- **Pythium** — warm-water root-rot pathogen; the dominant disease pressure in any reservoir above 72°F. Why reservoir-temperature management is the #1 Dragoon variable.

**The single canonical rule for this guide.** Masterblend pepper feeding at fruit sizing uses **2.4 g/gal Epsom salt** — double the standard fruiting Mg rate. This applies whether the magnesium is purchased Epsom salt or foraged

*epsomite* from a Sulphur Springs Valley seep. The rule is named in Guide 2 (Peppers) and Guide 3 (Bato Buckets) and lives here, in the nutrient program where it is actually dosed.

#### LOCAL TIP

**Where the boundary with Guide 3 sits.** Water treatment — the 5-step Cochise bicarbonate scrub, the choice of source water, the reservoir build — lives in Guide 3 §4 and is referenced briefly here when it affects nutrient mixing. The system build (buckets, drain channel, pump sizing, daily chores) lives in Guide 3 §2 and §3. This guide owns the **chemistry** that goes into that build.

## Part 1 — Masterblend

*The purchased precision track · the canonical 3-part recipe*

### 1.1 Why Masterblend 4-18-38

*Masterblend 4-18-38* is a professional-grade, water-soluble fertiliser used by commercial greenhouse growers worldwide. Combined with calcium nitrate and magnesium sulphate (Epsom salt), it delivers every essential macro- and micronutrient in precise, plant-available form. It is the canonical nutrient program for this library and the reason the rest of the guides can talk about “feeding the buckets” without ambiguity.

The three-part structure exists for a non-negotiable chemical reason: **calcium nitrate cannot share a concentrate with phosphate or sulphate**. Mix them together in concentrated form and they react instantly to form calcium phosphate and calcium sulphate precipitates — fluffy white solids that drop out of solution and lock the nutrients away from the plants. The split is not arbitrary; it is what keeps the chemistry stable.

| Why this program, for the Dragoons                       | What it solves                                                                                                  |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Hard SE Arizona well water already contributes Ca and Mg | A 3-part split lets you reduce Cal-Nitrate or Epsom independently to compensate, without throwing off the rest. |
| Soft monsoon rainwater contributes almost nothing        | Same split lets you run the full standard rate on rainwater without surprises.                                  |
| Wide temperature swings shift nutrient demand            | Adjusting <i>concentration</i> (the rate) is easy; adjusting <i>ratio</i> is rare.                              |
| Multiple crops on one reservoir is common                | Pepper, tomato, and basil all run on the same three jars, just at different weights.                            |

### The 3-part system

| Part | Product                         | NPK      | Primary role                           | Mix order        |
|------|---------------------------------|----------|----------------------------------------|------------------|
| A    | <i>Masterblend 4-18-38</i>      | 4-18-38  | P, K, Mg + all micronutrients          | 1st              |
| B    | Magnesium sulphate (Epsom salt) | 0-0-0    | Mg, S — boosts Masterblend Mg          | 2nd              |
| C    | Calcium nitrate 15.5-0-0        | 15.5-0-0 | Ca, N — never mix with A+B concentrate | 3rd, always last |



**WARNING**

Never mix calcium nitrate directly with *Masterblend* concentrate or with an A+B concentrate. They react and precipitate, locking out nutrients. Dissolve each part separately into the reservoir, or keep separate A and B stock solutions and a separate C stock solution.

**1.2 Stock solution preparation**

For larger systems or repeated mixing, stock concentrates save time and improve consistency. Keep Stock A and Stock B separate from Stock C — never combine them as concentrates.

| Stock   | Contents                                      | Concentration                       | Dose rate                       |
|---------|-----------------------------------------------|-------------------------------------|---------------------------------|
| Stock A | <i>Masterblend</i><br>4-18-38 + Epsom<br>salt | 1 lb of each per<br>gallon of water | 1 oz per gallon of<br>reservoir |
| Stock B | Calcium nitrate<br>15.5-0-0 only              | 1 lb per gallon of<br>water         | 1 oz per gallon of<br>reservoir |

**LOCAL TIP**

Label stock bottles with the date mixed. Use within two weeks — *Masterblend* concentrates begin to drop their micronutrient chelates after that. Store in a cool, dark spot to slow degradation. Use opaque HDPE bottles; clear glass on a sunny shelf will halve the shelf life.

**Stock-vs-direct decision**

| You should use stocks if...                         | You should mix direct if...                 |
|-----------------------------------------------------|---------------------------------------------|
| You have 4+ buckets and refill the reservoir weekly | You have a single DWC bucket or small bench |
| You mix the same recipe every week                  | You change recipe between crops often       |
| You travel and need a helper to mix                 | You enjoy weighing every batch by hand      |
| You want repeatability across years                 | You are still learning the program          |

**1.3 Mixing order — the strict A → Epsom → CaN-last rule**

This is the most important procedural rule in Part 1. Order matters because of the precipitation chemistry described above.

| Step | Action                                                 | What to observe                                                       |
|------|--------------------------------------------------------|-----------------------------------------------------------------------|
| 1    | Fill reservoir with pre-treated water (see Guide 3 §4) | pH 5.8–6.0, fizzing stopped                                           |
| 2    | Dissolve <i>Masterblend 4-18-38</i> fully              | Solution turns pale green; should be clear, not cloudy                |
| 3    | Dissolve Epsom salt fully                              | Solution remains clear                                                |
| 4    | Dissolve calcium nitrate fully <b>last</b>             | Solution stays clear — any cloudiness means precipitation has started |
| 5    | Final pH and EC check                                  | Adjust pH to 5.8–6.2; record EC                                       |

If the solution turns cloudy at step 4, you have either skipped the bicarbonate scrub (Guide 3 §4) or added calcium nitrate before the other two were fully dissolved. Pour off the cloudy reservoir, rinse, and start over with cleaner water. Do not “fix” a cloudy reservoir — the precipitated Ca and P are gone from solution.

## 1.4 Per-gallon feeding charts by crop

The base ratio is 2:1:2 (*Masterblend* : Epsom : Cal-Nitrate). Adjust *concentration*, not *ratio*, with one canonical exception: peppers at fruiting take a magnesium bump (see §1.4.2).

### 1.4.1 Tomato — full cycle

| Week          | Stage                 | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|---------------|-----------------------|------------|---------------|---------------|-----------|---------|
| 1–2           | Germination           | —          | —             | —             | 0.0       | 6.0     |
| 2–4           | Seedling              | 1.2        | 0.6           | 1.2           | 0.8–1.0   | 5.8–6.0 |
| 4–7           | Early veg             | 2.4        | 1.2           | 2.4           | 1.4–1.8   | 5.8–6.0 |
| 7–10          | Late veg / transition | 2.8        | 1.4           | 2.8           | 1.8–2.2   | 5.8–6.2 |
| 10–14         | First truss flowering | 3.0        | 1.5           | 3.0           | 2.0–2.4   | 5.8–6.2 |
| 14–20         | Heavy fruiting        | 3.6        | 1.8           | 3.6           | 2.4–3.0   | 6.0–6.2 |
| Final 1–2 wks | Flush / finish        | 0.5        | 0.3           | 0.5           | 0.5–0.8   | 6.0     |

### 1.4.2 Pepper — full cycle (with canonical Mg bump)

| Week | Stage                     | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|------|---------------------------|------------|---------------|---------------|-----------|---------|
| 1-3  | Seedling                  | 1.2        | 0.6           | 1.2           | 0.8-1.0   | 5.8-6.0 |
| 3-6  | Vegetative                | 2.4        | 1.2           | 2.4           | 1.4-1.8   | 5.8-6.0 |
| 6-10 | Pre-flower                | 2.8        | 1.4           | 2.8           | 1.8-2.2   | 5.8-6.2 |
| 10+  | <b>Fruiting (Mg bump)</b> | 3.0        | <b>2.4</b>    | 3.0           | 2.2-2.8   | 6.0-6.2 |

**Pepper fruiting Epsom: 2.4 g/gal, not 1.5.** Peppers lift more magnesium through the fruiting phase than any other common hydroponic crop. Hold the Mg bump from first fruit set through harvest. This value supersedes the 1.5 g/gal figure that appears in some upstream Masterblend reference tables — those tables average tomatoes and peppers and undercount peppers. For tomatoes the Mg bump is smaller, 1.5-1.8 g/gal, and stays on the row 14-20 of the tomato table above. This conflict resolution is one of the reasons this consolidated guide exists.

### 1.4.3 Leafy greens & herbs

| Crop     | Stage      | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|----------|------------|------------|---------------|---------------|-----------|---------|
| Lettuce  | All stages | 1.2-1.8    | 0.6-0.9       | 1.2-1.8       | 0.8-1.4   | 5.8-6.2 |
| Basil    | Vegetative | 2.0        | 1.0           | 2.0           | 1.4-1.8   | 5.8-6.0 |
| Kale     | All stages | 2.0-2.4    | 1.0-1.2       | 2.0-2.4       | 1.4-2.0   | 5.8-6.2 |
| Spinach  | All stages | 1.6        | 0.8           | 1.6           | 1.0-1.6   | 6.0-6.5 |
| Cilantro | All stages | 1.2-1.6    | 0.6-0.8       | 1.2-1.6       | 0.8-1.4   | 6.0-6.5 |

### 1.4.4 5-gallon batch reference

| Use case                   | Masterblend | Epsom       | Cal-Nitrate | Target EC |
|----------------------------|-------------|-------------|-------------|-----------|
| Seedlings / cuttings       | 6 g         | 3 g         | 6 g         | 0.8-1.0   |
| Vegetative                 | 12 g        | 6 g         | 12 g        | 1.4-1.8   |
| Fruiting (pepper, Mg bump) | 15 g        | <b>12 g</b> | 15 g        | 2.2-2.8   |
| Fruiting (tomato)          | 15 g        | 7.5 g       | 15 g        | 2.0-2.4   |
| Heavy fruiting (tomato)    | 18 g        | 9 g         | 18 g        | 2.4-3.0   |

**LOCAL TIP**

A digital scale that reads to 0.1 g is non-negotiable for this program. The difference between 1.2 g and 2.4 g of Epsom on a single gallon is the difference between healthy mid-leaf colour and interveinal yellowing on the lower trusses. A \$15 jeweller's scale from ACE Hardware in Sierra Vista pays for itself in saved foliage.

**1.5 Full nutrient profile at standard rate**

This is what 2.4 g/1.2 g/2.4 g per gallon (the standard vegetative dose) delivers, ppm by ppm.

| Macro          | ppm | Micro           | ppm  |
|----------------|-----|-----------------|------|
| Nitrogen (N)   | 116 | Iron (Fe)       | 2.4  |
| Phosphorus (P) | 47  | Boron (B)       | 1.2  |
| Potassium (K)  | 188 | Manganese (Mn)  | 1.2  |
| Calcium (Ca)   | 113 | Zinc (Zn)       | 0.3  |
| Magnesium (Mg) | 41  | Copper (Cu)     | 0.3  |
| Sulfur (S)     | 58  | Molybdenum (Mo) | 0.06 |
| Chloride (Cl)  | 12  |                 |      |

This profile is the target Part 2 (DIY) tries to hit with locally-sourced equivalents.

**1.6 EC and pH daily protocol**

Consistent monitoring is what separates good results from great ones. With *Masterblend* and hard desert water, daily checks for the first two weeks of any new reservoir will reveal your water's behaviour and let you build a predictable maintenance routine.

| Frequency | Task     | Notes                                                                                                   |
|-----------|----------|---------------------------------------------------------------------------------------------------------|
| Morning   | pH check | Adjust if outside 5.8-6.2. Note drift direction and amount — patterns matter more than single readings. |
| Morning   | EC check | Top off with half-strength mix if EC dropped from evaporation, plain water if EC rose.                  |

| Frequency | Task                  | Notes                                                                    |
|-----------|-----------------------|--------------------------------------------------------------------------|
| Morning   | Water level           | Top off reservoirs. Never let roots dry out.                             |
| Evening   | Visual plant check    | Leaf colour, wilting, pests, root colour through net pot.                |
| Weekly    | Full reservoir change | Fruiting: every 7–10 days.<br>Greens: every 14 days.                     |
| Weekly    | Meter calibration     | pH meter with pH 4.0 and 7.0 buffer. EC meter with 1.413 mS/cm standard. |
| Weekly    | Root inspection       | Healthy = bright white.<br>Brown/slippy = root rot — see Guide 3 §7.     |
| Monthly   | System flush          | Flush lines and reservoir with plain water between cycles.               |

### Understanding EC drift

| Observation           | Cause                                                                 | Action                                                        |
|-----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------|
| EC rising             | Plants drinking water faster than nutrients                           | Top off with plain RO/rainwater, not full-strength mix        |
| EC falling            | Plants consuming nutrients faster than water                          | Top off with half-strength <i>Masterblend</i> mix             |
| EC stable, pH rising  | Classic bicarbonate effect from hard water                            | Pre-treat next reservoir before adding nutrients (Guide 3 §4) |
| EC stable, pH falling | Biological activity or $\text{NH}_4^+$ uptake, common in warm weather | Adjust up with pH Up — see Part 3                             |

### 1.7 Deficiency diagnosis table

This is your single-page diagnostic when leaves change colour. Read symptom first, then likely deficiency, then the *Masterblend* fix.

| Symptom                                        | Likely deficiency               | Masterblend fix                                               | pH range cause                                 |
|------------------------------------------------|---------------------------------|---------------------------------------------------------------|------------------------------------------------|
| Yellow between leaf veins — older leaves first | Magnesium (Mg)                  | Double Epsom dose (2.4 g/gal) for 1-2 weeks                   | Locked out below 5.5                           |
| Yellow between leaf veins — new leaves first   | Iron (Fe) or Manganese (Mn)     | Check pH — most common cause. Adjust to 5.8-6.0               | Locked out above 6.5                           |
| Brown / dark tips, necrotic leaf edges         | Calcium (Ca) or Potassium (K)   | Check Cal-Nitrate was mixed last. Increase to 3.0 g/gal       | Locked out below 5.5 or above 7.0              |
| Blossom end rot on tomatoes / peppers          | Calcium deficiency              | Raise Cal-Nitrate to 3.6 g/gal. Check EC isn't too high       | pH out of range or EC too high blocking uptake |
| Purple / reddish leaf undersides               | Phosphorus (P)                  | Check pH — P locks out below 5.0. Warm to 70 °F+              | Locked out below 5.0 or above 7.5              |
| Overall pale, slow growth                      | Nitrogen (N) — too low EC       | Increase overall dose by 0.5 g/gal                            | Check EC < 1.0 for fruiting crops              |
| Dark green leaves, stocky but slow             | Nitrogen — too high EC          | Dilute reservoir. Partial change with fresh water             | EC above 3.5 in most cases                     |
| Cupped new leaves, sticky residue              | Boron (B) excess                | Flush with plain water. Reduce Masterblend by 25 % next batch | Often pH-driven — check below 5.5              |
| Leaf tip burn (lettuce, basil)                 | Salt accumulation / EC too high | Dilute or change reservoir                                    | EC > 2.0 in greens                             |

### LOCAL TIP

**The pepper interveinal yellow on the bottom-third leaves.** This is the canonical Dragoons deficiency: monsoon rainwater is soft and contributes almost no Mg, peppers are heavy Mg users, and the bottom leaves are the first to mobilise Mg out. Confirm by checking pH (should be 5.8–6.2, not lower) and EC (should be in range), then run the 2.4 g/gal Epsom bump for two weeks before changing anything else.

## 1.8 Crop-specific optimisation

Quick-reference parameter cards. These pull together EC, pH, light, temperature, and VPD targets for the four most common Dragoons crops. PPFD/DLI/VPD numbers

are detailed in Part 4.

### Indeterminate tomato

| Parameter      | Setting                                                                                      |
|----------------|----------------------------------------------------------------------------------------------|
| System         | Bato Bucket (perlite/coco) or DWC                                                            |
| EC             | 2.0–3.0 mS/cm                                                                                |
| pH             | 5.8–6.2                                                                                      |
| Light (PPFD)   | 600–900 $\mu\text{mol}/\text{m}^2/\text{s}$                                                  |
| DLI            | 25–35 $\text{mol}/\text{m}^2/\text{day}$                                                     |
| Temp day/night | 75–82 °F / 62–68 °F                                                                          |
| Photoperiod    | 14–16 hrs                                                                                    |
| VPD fruiting   | 1.2–1.6 kPa                                                                                  |
| Key tip        | Calcium is critical — never skip Cal-Nitrate. Blossom end rot = Ca deficiency or pH lockout. |

### Chile & bell pepper

| Parameter      | Setting                                                                                     |
|----------------|---------------------------------------------------------------------------------------------|
| System         | Bato Bucket or DWC                                                                          |
| EC             | 1.8–2.8 mS/cm                                                                               |
| pH             | 5.8–6.2                                                                                     |
| Light (PPFD)   | 600–800 $\mu\text{mol}/\text{m}^2/\text{s}$                                                 |
| DLI            | 22–30 $\text{mol}/\text{m}^2/\text{day}$                                                    |
| Temp day/night | 72–82 °F / 60–65 °F                                                                         |
| Photoperiod    | 14–16 hrs                                                                                   |
| VPD fruiting   | 1.2–1.6 kPa                                                                                 |
| Key tip        | Hold Epsom at <b>2.4 g/gal</b> from first fruit set through harvest. The canonical Mg bump. |

### Lettuce & leafy greens

| Parameter | Setting       |
|-----------|---------------|
| System    | NFT or Kratky |
| EC        | 0.8–1.6 mS/cm |

| Parameter    | Setting                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| pH           | 5.8-6.5                                                                                                                           |
| Light (PPFD) | 200-400 $\mu\text{mol}/\text{m}^2/\text{s}$                                                                                       |
| DLI          | 12-17 $\text{mol}/\text{m}^2/\text{day}$                                                                                          |
| Temp         | 60-72 °F — cool is better                                                                                                         |
| Photoperiod  | 16-18 hrs                                                                                                                         |
| VPD          | 0.8-1.2 kPa                                                                                                                       |
| Key tip      | Lower EC produces tender, mild leaves. High EC causes tip burn and bitterness. In summer, grow in the coolest part of your space. |

## Basil

| Parameter    | Setting                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| System       | DWC or NFT                                                                                                                                |
| EC           | 1.4-2.0 mS/cm                                                                                                                             |
| pH           | 5.8-6.2                                                                                                                                   |
| Light (PPFD) | 300-500 $\mu\text{mol}/\text{m}^2/\text{s}$                                                                                               |
| DLI          | 15-20 $\text{mol}/\text{m}^2/\text{day}$                                                                                                  |
| Temp         | 70-85 °F — loves heat                                                                                                                     |
| Photoperiod  | 16-18 hrs                                                                                                                                 |
| VPD          | 1.0-1.4 kPa                                                                                                                               |
| Key tip      | Harvest frequently to prevent bolting. High PPFD produces more essential oils and stronger flavour. An ideal Dragoon monsoon-season crop. |

### LOCAL TIP

**Cross-reference.** Heritage Andean peppers (Aji, Rocoto, Manzano) and why the Dragoons' elevation suits them are covered in Guide 2. Rocoto's known weak points in recirculating systems (cooler root-zone requirement, sensitivity to EC spikes) are also flagged there. This guide is the home for the nutrient program; that guide is the home for the crop biology.



## Part 2 — Locally-sourced DIY

*The off-grid track · oak ash, bat guano, caliche, epsomite · same ppm targets*

### 2.1 Why this part exists

Cochise County is one of the richest mineral landscapes in the American Southwest. Oak for potassium, bat caves for phosphorus, caliche for calcium, *epsomite* for magnesium, Bisbee copper and Tombstone zinc for trace elements — the same geology that made this a mining frontier makes it an unusually self-sufficient place to grow food hydroponically.

The purchased *Masterblend* program in Part 1 is faster and more precise. The locally-sourced program here takes longer, drifts more, and asks more of your measuring kit — but it costs almost nothing to run, and it ties the reservoir back into the land it sits on. Either gets a pepper to ripen. Most growers end up somewhere in the middle: a bag of *Masterblend* for the base, and a brew of caliche calcium or an *epsomite* crust for the parts the valley gives you for free.

Everything in Part 2 targets the same nutrient ppm profile as Part 1: **116 N / 47 P / 188 K / 113 Ca / 41 Mg / 58 S**. The difference is where the raw materials come from. You can run the locally-sourced program end-to-end, or you can blend — for example, purchased *Masterblend* + local caliche calcium + local *epsomite*. The crop ppm targets do not change either way.

### 2.2 Target ppm profile for the three-part replacement

This is the same profile as Part 1 §1.5, repeated here so Part 2 can be read standalone.

| Macronutrient  | Target ppm | Micronutrient       | Target ppm |
|----------------|------------|---------------------|------------|
| Nitrogen (N)   | 116        | Iron (Fe, chelated) | 2.4        |
| Phosphorus (P) | 47         | Boron (B)           | 1.2        |
| Potassium (K)  | 188        | Manganese (Mn)      | 1.2        |
| Calcium (Ca)   | 113        | Zinc (Zn)           | 0.3        |
| Magnesium (Mg) | 41         | Copper (Cu)         | 0.3        |
| Sulfur (S)     | 58         | Molybdenum (Mo)     | 0.06       |

### The three parts, mapped to local sources

| Part                         | Purchased (Part 1)                        | Local equivalent                                                                   |
|------------------------------|-------------------------------------------|------------------------------------------------------------------------------------|
| A — K, P, N + micronutrients | <i>Masterblend 4-18-38</i> (12 g / 5 gal) | Oak-ash potash + aerated bat-guano tea + chelated iron + micronutrient micro-doses |
| B — Mg + S                   | Epsom salt (6 g / 5 gal)                  | <i>Epsomite</i> crust from Sulphur Springs Valley                                  |
| C — Ca + N                   | Calcium nitrate (12 g / 5 gal)            | Caliche dissolved in apple-cider vinegar + nitrified guano tea                     |

**WARNING**

Never pre-combine the local Parts A, B, and C as concentrates. They will react and precipitate exactly the same way the purchased equivalents do — calcium phosphate and calcium sulphate are calcium phosphate and calcium sulphate regardless of source. Dissolve each separately into the reservoir.

## 2.3 Local sources

### 2.3.1 Potassium from oak ash — Cochise County oak

Oak woodland is the Dragoons' signature plant community. *Quercus emoryi* (Emory oak) and *Quercus arizonica* (Arizona white oak) dominate the foothill slopes between 4,500 and 6,500 ft; *Quercus hypoleucoides* (silverleaf oak) and *Q. grisea* (gray oak) come in higher. All four work for ash; Emory is the most abundant and the easiest to harvest from windfall.

| Step | Action                                               | Notes                                                                 |
|------|------------------------------------------------------|-----------------------------------------------------------------------|
| 1    | Burn seasoned oak hardwood completely                | No charcoal chunks, no resinous wood, no pine. White-gray ash only.   |
| 2    | Pack a non-metallic container loosely with cold ash  | Food-grade bucket or large glass jar.                                 |
| 3    | Steep in 2-3 volumes of rainwater (or RO) for 24 hrs | Do <b>not</b> use well water — bicarbonates interfere with the leach. |
| 4    | Strain through fine cloth into a dark glass bottle   | The liquid is your potash extract.                                    |

| Step | Action                        | Notes                          |
|------|-------------------------------|--------------------------------|
| 5    | Test pH (expect 11-13) and EC | Label, date, and store sealed. |

**LOCAL TIP**

**Emory oak in the foothills.** *Q. emoryi* is the dominant oak below 6,000 ft on the east and west slopes of the Dragoons, and a Sulphur Springs Valley fireplace will run on Emory windfall through most of the winter. The ash is among the highest in soluble potassium of any North American oak — about 6-10 % K<sub>2</sub>O by mass after a clean burn. One winter's fireplace ash will supply potash for a 20-bucket bato system for a full season.

**WARNING**

Oak-ash potash extract is strongly alkaline (pH 11-13) and caustic. Handle with gloves and eye protection. This is the same material as the pH-up base in Part 3 — see §3.3.1 for the safety detail.

**2.3.2 Nitrogen from bat guano tea — Kartchner and local caves**

The Whetstones (just north of the Dragoons) house Kartchner Caverns State Park, and the broader Cochise County limestone karst has dozens of smaller wild caves and old mine adits used as roosts by *Tadarida brasiliensis* (Mexican free-tailed bat) and *Antrozous pallidus* (pallid bat). Bat guano is high in both phosphorus and nitrogen, and when aerated as a tea, the nitrogen converts from ammonium to plant-available nitrate.

| Step | Action                                | Notes                                                                                                            |
|------|---------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1    | Collect bat guano from a legal source | <b>Not from Kartchner itself</b> — it is a state park. Local mine adits and private caves with owner permission. |
| 2    | Steep 1 cup guano per gallon of water | Use rainwater or RO; well-water bicarbonates buffer the chemistry.                                               |

| Step | Action                                            | Notes                                                                                                                            |
|------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 3    | Aerate vigorously for 24–48 hrs                   | An aquarium air pump with two airstones is enough. Aeration converts $\text{NH}_4^+$ to $\text{NO}_3^-$ via nitrifying bacteria. |
| 4    | Strain finely through cloth, then a coffee filter | The clear amber liquid is your N + P source.                                                                                     |
| 5    | Use within 48 hours of aeration completing        | After that the microbial population shifts and you lose nitrate.                                                                 |

**WARNING**

**Histoplasmosis risk.** Bat guano can carry *Histoplasma capsulatum* spores. Harvest in a N95 mask, do not stir dry guano in enclosed spaces, and wash hands and clothing afterward. This is a real (though small) Cochise County exposure risk — take it seriously.

**LOCAL TIP**

A few abandoned mine workings in the Dragoons foothills are well-known bat roosts and are off-limits during summer pup season (June–August). Harvest in winter only, take small amounts, and never disturb a roosting colony. The fertiliser is not worth a colony.

**2.3.3 Calcium from caliche — the Dragoons hardpan**

Caliche — calcium carbonate hardpan — is ubiquitous throughout the Sulphur Springs Valley east of the Dragoons and forms a continuous layer under most of the Dragoons' alluvial soils. It is the same material as agricultural lime, and it dissolves readily in dilute acid to form plant-available calcium.

| Step | Action                                             | Notes                                          |
|------|----------------------------------------------------|------------------------------------------------|
| 1    | Collect caliche chunks from exposed hardpan layers | Road cuts, eroded arroyos, garden excavations. |
| 2    | Crush to gravel size                               | Hammer on a hard surface; wear safety glasses. |

| Step | Action                                                   | Notes                                                                                              |
|------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 3    | Dissolve slowly in apple-cider vinegar                   | Produces calcium acetate, fully plant-available. Expect fizzing — that is CO <sub>2</sub> leaving. |
| 4    | Wait for fizzing to stop completely                      | Filter through cloth. The clear liquid is your Ca source.                                          |
| 5    | Combine with nitrified guano tea                         | Provides Ca and N together — same pairing as purchased calcium nitrate.                            |
| 6    | Alternative: dolomite limestone dissolved in dilute acid | Provides both Ca and Mg simultaneously.                                                            |

**LOCAL TIP**

**Why caliche, not eggshells.** Eggshells are calcium carbonate too, but the surface area is wrong — they dissolve over weeks, not hours. Caliche fragments at gravel size give you usable calcium acetate in a single afternoon. The Sulphur Springs Valley has an effectively infinite supply.

**2.3.4 Magnesium and sulfur from Sulphur Springs Valley *epsomite***

The Sulphur Springs Valley is named for its mineral deposits. Look for *epsomite* — white crystalline crusts around dry springs, mineral seeps, and the margins of playas. *Epsomite* is magnesium sulphate heptahydrate in its natural form, chemically identical to commercial Epsom salt.

| Step | Action                                           | Notes                                                             |
|------|--------------------------------------------------|-------------------------------------------------------------------|
| 1    | Scout dry seeps, spring margins, and playa edges | Late spring (after the dry winter) is the best collecting window. |
| 2    | Collect white crystalline crust                  | A clean trowel and a glass jar. Avoid darker mineral mixes.       |
| 3    | Dissolve directly in reservoir water             | Same dose rates as commercial Epsom salt — see §1.4.              |

**Alternative: dissolve dolomite in sulphur-spring water**

If you cannot find *epsomite*, dissolving dolomite limestone ( $\text{CaMg}(\text{CO}_3)_2$ ) in sulphur-spring water gives you Ca and Mg together with some native sulphate. This is messier and harder to dose accurately, but workable.

**The pepper Mg bump applies here too.** Whether the magnesium is purchased Epsom salt or foraged *epsomite*, peppers at fruiting want **2.4 g-equivalent per gallon** — the canonical bump from §1.4.2. The chemistry is the same; only the supply chain is different.

**2.4 Micronutrient sourcing**

The macros are easy; the micros are where local sourcing gets ambitious. Most growers eventually buy a small bottle of chelated micronutrient mix and call it done.

| Nutrient        | Local source                                                                                | Method                                                                                                                          |
|-----------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Iron (Fe)       | Red granite clay + oak bark / leaves                                                        | Brew strong oak-bark tea. Soak iron-rich red clay in it 48 hrs. Strain. The tannins chelate the iron into plant-available form. |
| Manganese (Mn)  | Dark metamorphic rock, streambed sediment                                                   | Collect mineral-stained streambed mud. Brew as dilute tea. Strain before use.                                                   |
| Zinc (Zn)       | Tombstone / Bisbee zinc-bearing ores                                                        | Dissolve a small zinc-ore chip in dilute acid. Heavily dilute before adding to reservoir.                                       |
| Copper (Cu)     | Bisbee copper ores ( <i>malachite</i> , <i>azurite</i> )                                    | Dissolve a small piece in dilute apple-cider vinegar. Tiny amounts only — Cu is toxic at high doses.                            |
| Boron (B)       | Desert caliche, dry lakebeds, borax deposits                                                | Use Sulphur Springs Valley soil tea. Use very sparingly — toxic above 2 ppm.                                                    |
| Molybdenum (Mo) | Wild legume root nodules ( <i>mesquite</i> , <i>Prosopis velutina</i> , palo verde, clover) | Ferment root nodules in water. Provides trace Mo — the hardest element to source locally.                                       |

**WARNING**

**Micronutrient toxicity.** Copper, boron, and zinc are toxic to plants at high concentrations — the window between deficient and toxic for B is narrower than for any other nutrient. Always make a very dilute stock solution (1:100 or more) and add drop by drop, testing EC between additions. Zinc and manganese follow the same rule — small amounts, dilute, measure.

**2.5 Feasibility scoring**

A practical look at how achievable each nutrient is locally. The micros mark where most growers compromise.

| Component               | Locally achievable? | Notes                                                            |
|-------------------------|---------------------|------------------------------------------------------------------|
| Potassium (K)           | Yes                 | Oak wood ash is an excellent source.                             |
| Phosphorus (P)          | Yes                 | Bat guano from Cochise County caves and mine adits.              |
| Calcium (Ca)            | Yes                 | Caliche is ubiquitous.                                           |
| Magnesium (Mg)          | Likely              | Hunt for <i>epsomite</i> deposits in the Sulphur Springs Valley. |
| Sulfur (S)              | Yes                 | The valley is named for its sulphur deposits.                    |
| Iron (Fe), chelated     | Yes                 | Oak tannin + red clay method is effective.                       |
| Copper (Cu)             | Yes                 | Bisbee copper ores are world-class.                              |
| Boron (B)               | Yes                 | AZ desert soils concentrate boron naturally.                     |
| Nitrogen (nitrate form) | Partial             | Hard to get pure nitrate — aerated guano tea helps but drifts.   |
| Mn, Zn (trace)          | Difficult           | Available but hard to dose accurately.                           |
| Molybdenum (Mo)         | Very hard           | Legume tea only provides trace amounts.                          |

**LOCAL TIP**

**The pragmatic Dragoons split.** Use local Parts B and C (*epsomite*, caliche, guano, oak ash) and purchase a low-cost chelated micronutrient mix for the trace elements (Mo, Zn, Mn, B). You get most of the self-sufficiency at a fraction of the accuracy risk. A \$20 bottle of micronutrient mix lasts two years and weighs less than the avoided foraging trips.

**2.6 The 8-step DIY workflow**

This is the practical sequence for a grower running the full local program. Stage the brews in advance — each takes its own time window.

| Step | Action                                                | Source                                         |
|------|-------------------------------------------------------|------------------------------------------------|
| 1    | Brew oak-ash potash extract                           | Local oaks → burn → steep 24 hrs               |
| 2    | Prepare aerated bat-guano tea                         | Cochise County bat roosts → 24-48 hr aeration  |
| 3    | Make caliche-vinegar calcium extract                  | Sulphur Springs Valley caliche + cider vinegar |
| 4    | Brew oak-tannin + red-clay iron chelate               | Local granite hillsides → 48 hr steep          |
| 5    | Collect <i>epsomite</i> (or use purchased Epsom salt) | Local mineral seeps or buy                     |
| 6    | Make copper micro-dose                                | Bisbee <i>malachite</i> in dilute acid         |
| 7    | Brew desert-soil boron tea                            | Sulphur Springs caliche soil                   |
| 8    | Combine all parts separately into reservoir           | Check EC + pH after each addition              |

Regardless of how nutrients are sourced, an **EC meter and a pH meter are essential**. Target EC 1.8-2.4 mS/cm and pH 5.8-6.2 for most fruiting crops. These instruments cannot be replaced with local materials. Buy quality (Blue-lab, Apera, or Hanna) and calibrate weekly.



## Part 3 — pH adjustment recipes

*Acids and bases · foraged and purchased · daily-use tools*

### 3.1 Why pH control matters

Whether you're using SE Arizona well water or collected monsoon rainwater, pH management is the most important daily task in hydroponics. Nutrients become chemically unavailable outside the ideal range — plants can starve in a perfectly mixed reservoir if the pH is wrong.

| pH range         | What happens                                            | Common in the Dragoons?                     |
|------------------|---------------------------------------------------------|---------------------------------------------|
| Below 5.5        | Iron, manganese, copper toxicity. Root damage.          | Rare — only if overdosing acid              |
| 5.8–6.2 (target) | All nutrients fully available. Optimal for most crops.  | Requires treatment of local water           |
| 6.5–7.0          | Iron and manganese start locking out. Yellowing begins. | Common with well water — correct it         |
| 7.5–8.5          | Most micronutrients locked out. Serious deficiencies.   | Very common with untreated SE AZ well water |

### The bicarbonate problem in SE Arizona

SE Arizona well water is rich in bicarbonates — dissolved minerals that act as a pH buffer, constantly pushing water toward 7.5–8.5 even after you correct it. The fix is to neutralise the bicarbonates first, then fine-tune. You will see fizzing when you add acid — that is CO<sub>2</sub> releasing as the bicarbonates neutralise. **Wait for the fizzing to stop before a final adjustment.**

**Cross-reference.** The full **5-step Cochise bicarbonate water-treatment protocol** is in **Guide 3 §4** — it is the canonical home for water treatment in this library. Part 3 here owns the chemistry of the acids and bases themselves; Guide 3 owns the sequence of how they are applied to reservoir water.

### WARNING

Always adjust pH **after** mixing nutrients, not before. The nutrients themselves shift pH, and pre-adjusting wastes acid. Always **add acid to water — never water to acid**. Work in small amounts.

### 3.2 pH Down recipes

The Dragoons and surrounding Cochise County provide several natural acid sources. These are best for minor daily adjustments on small systems. They are free, renewable, and satisfying to produce from your own landscape. For consistency and precision, the purchased homemade options are the recommended daily-driver.

#### 3.2.1 Prickly-pear vinegar (foraged)

Prickly pear cactus (*Opuntia engelmannii*, *O. phaeacantha*) is abundant throughout the Dragoons and Sulphur Springs Valley. The fruit ferments readily into a mild fruit vinegar with a final pH of 3.0–3.5 — gentle enough for easy dosing on small systems.

**What you need:** ripe prickly-pear fruits (tuna) · a glass jar with a loose lid · a fine cloth strainer · a pH meter.

| Step | Action                                                    | Notes                                                               |
|------|-----------------------------------------------------------|---------------------------------------------------------------------|
| 1    | Harvest 10–15 ripe red/purple fruits using tongs          | Rinse spines off under water                                        |
| 2    | Mash thoroughly in a glass jar, including skins and seeds | They carry the wild yeasts                                          |
| 3    | Cover loosely with cloth (not airtight)                   | Leave at room temp 2–3 weeks, stirring daily                        |
| 4    | After 2 weeks, strain a small sample and test pH          | Target is 3.0–3.5                                                   |
| 5    | Strain through fine cloth into a clean glass jar          | Discard solids. Label and date.                                     |
| 6    | Refrigerate                                               | Lasts 3–6 months; use within 1 week once opened at room temperature |

| System size | Starting dose                     | Tip                                                        |
|-------------|-----------------------------------|------------------------------------------------------------|
| 1 gallon    | 1–2 ml, stir, wait 10 min, retest | Keep a small spray bottle for easy micro-dosing            |
| 3 gallons   | 3–5 ml, stir, wait 10 min, retest | Add slowly — prickly-pear vinegar is gentle but cumulative |
| 5 gallons   | 5–8 ml, stir, wait 10 min, retest | Multiple small doses beat one large one                    |

**LOCAL TIP**

Prickly-pear vinegar adds very small amounts of natural fruit acids and trace minerals to your reservoir. At the low doses used for pH adjustment, this has negligible effect on the nutrient profile. *Opuntia engelmannii* is the canonical Dragoons species — harvest in September when the tunas are deep red.

**3.2.2 Fermented citric acid (foraged from local fruit or honey)**

Citric acid can be produced by fermenting sugar with *Aspergillus niger* mould, which grows naturally on overripe fruit skins in this region. This is, in fact, how industrial citric acid is manufactured — you are doing it at small scale. The result is stronger than vinegar and more precise to dose.

| Step | Action                                                                | Notes                                      |
|------|-----------------------------------------------------------------------|--------------------------------------------|
| 1    | Dissolve 2 tbsp local honey or prickly-pear juice in 1 cup warm water | Wide-mouth glass jar                       |
| 2    | Add a small piece of overripe citrus peel or lemon skin               | The white fuzz is <i>Aspergillus niger</i> |
| 3    | Cover with breathable cloth. Leave at 75–85 °F for 7–10 days          | A white/grey mould mat will form on top    |
| 4    | Test pH of the liquid beneath the mat                                 | Target is 2.0–3.0                          |
| 5    | Remove the mat and strain liquid through cloth                        | The clear acidic liquid is your pH down    |
| 6    | Dilute 1:4 with water before use                                      | Easier dosing                              |

**WARNING**

Do **not** consume the fermented citric-acid liquid — it is for plant use only. The *Aspergillus* mat is harmless in this context but should be discarded, not eaten. Work outdoors or with ventilation.

**3.2.3 Sulfur-dioxide acid water (foraged, advanced)**

The Sulphur Springs Valley east of the Dragoons contains native sulphur deposits. Burning sulphur produces sulphur-dioxide gas, which dissolves in water to form sulphurous acid — a strong, effective pH down. This method is powerful and best reserved for pre-treating large volumes of hard well water.

| Step | Action                                      | Notes                                                |
|------|---------------------------------------------|------------------------------------------------------|
| 1    | Gather native sulphur from mineral deposits | Yellow crystalline material from fumarolic areas     |
| 2    | Work completely outdoors, wind at your back | Never indoors — fumes are irritating                 |
| 3    | Ignite a small piece in a metal dish        | Burns with a blue flame and white fumes              |
| 4    | Hold a bucket of water downwind of the burn | Fumes dissolve into the surface                      |
| 5    | Test the water pH every few minutes         | Stop at pH 4.0–5.0                                   |
| 6    | Use immediately                             | Sulphur-acid water is unstable — use within 24 hours |

**LOCAL TIP**

Best use for this method: pre-treating a full reservoir of SE Arizona well water before adding nutrients. The strong acid neutralises bicarbonates efficiently. Then fine-tune with prickly-pear vinegar or citric-acid stock for the final pH adjustment.

**3.2.4 Citric-acid stock solution (purchased, best overall)**

Food-grade citric-acid powder is cheap (~\$5/lb), extremely stable, and makes a consistent pH down that closely mirrors commercial products. It does not add harmful ions to the reservoir and is the top recommendation for small hobby systems. A single pound will last a small grower 1–2 years.

| Step | Action                                                   | Notes                                                           |
|------|----------------------------------------------------------|-----------------------------------------------------------------|
| 1    | Buy food-grade citric-acid powder                        | Not industrial. Grocery stores, Amazon, homebrew suppliers.     |
| 2    | Dissolve 1 tbsp (≈12 g) in 1 cup (240 ml) of clean water | Stir until fully clear                                          |
| 3    | Pour into a dark glass bottle                            | Label: pH Down — Citric Acid Stock. Room temp, away from light. |
| 4    | Dose 1 ml at a time                                      | Stir, wait 5–10 min, retest before adding more                  |

| System size | Starting dose          | For bicarbonate pre-treatment                      |
|-------------|------------------------|----------------------------------------------------|
| 1 gallon    | 0.5–1 ml stock, retest | 3–5 ml stock until fizzing stops, then fine-tune   |
| 3 gallons   | 1–2 ml stock, retest   | 8–12 ml stock until fizzing stops, then fine-tune  |
| 5 gallons   | 2–3 ml stock, retest   | 12–20 ml stock until fizzing stops, then fine-tune |

**LOCAL TIP**

Citric acid breaks down naturally in the reservoir over 3–5 days via microbial activity, so it does not accumulate. It leaves no residual ions that would affect the nutrient balance. Safe, clean, effective — and the canonical daily-driver pH-down for this library.

**3.2.5 White vinegar stock (purchased, budget)**

Standard white vinegar (5 % acetic acid) is available everywhere and works as a mild, easy-to-dose pH down. The main downside is that acetate ions can accumulate in recirculating systems over time. For small systems with regular reservoir changes every 7–14 days, this is not a significant issue.

| Step | Action                                                                           |
|------|----------------------------------------------------------------------------------|
| 1    | Purchase standard white distilled vinegar, 5 % acidity. Any grocery brand.       |
| 2    | Use undiluted — at 5 % it is already dilute enough for easy dosing.              |
| 3    | Add 2–5 ml at a time. Stir, wait 10 min, retest.                                 |
| 4    | Change reservoir every 7–10 days for fruiting crops to prevent acetate build-up. |

| System size | Starting dose        | Notes                                          |
|-------------|----------------------|------------------------------------------------|
| 1 gallon    | 2–3 ml, stir, retest | Gentle — easier to control than stronger acids |
| 3 gallons   | 5–8 ml, stir, retest | Multiple small additions beat one large dose   |

| System size | Starting dose         | Notes                                            |
|-------------|-----------------------|--------------------------------------------------|
| 5 gallons   | 8-12 ml, stir, retest | Well-water pre-treatment may need 20-40 ml total |

**WARNING**

**Avoid apple-cider vinegar in the reservoir.** It contains organic compounds and sugars that promote bacterial growth and biofilm. Use white distilled vinegar only for reservoir pH adjustment. (Apple-cider vinegar is fine for the caliche-dissolving step in Part 2 — different application, different chemistry.)

**3.2.6 Diluted phosphoric acid (purchased, closest to commercial)**

Commercial pH Down products are typically 85 % phosphoric acid diluted to around 10-20 % concentration. You can buy food-grade phosphoric acid and make the same product at a fraction of the cost. Phosphoric acid also adds a small amount of phosphorus to the reservoir — a benefit for fruiting crops.

| Step | Action                                                                    | Notes                                                                    |
|------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1    | Buy food-grade phosphoric acid 85 %                                       | Used in cola production. Homebrew and hydroponic suppliers online.       |
| 2    | Gloves and eye protection                                                 | The 85 % solution is a strong acid                                       |
| 3    | Add 1 part phosphoric acid to 9 parts water, slowly — never water to acid | ~8.5 % working solution                                                  |
| 4    | Label clearly                                                             | pH Down — Phosphoric 8.5 % — HANDLE WITH CARE. Store away from children. |
| 5    | Dose 0.5 ml at a time                                                     | Stronger than vinegar or citric acid — very small doses                  |

| System size | Starting dose            | Notes                                         |
|-------------|--------------------------|-----------------------------------------------|
| 1 gallon    | 0.2-0.5 ml, stir, retest | Very small doses — much stronger than vinegar |
| 3 gallons   | 0.5-1.0 ml, stir, retest | Drop by drop with a syringe for best control  |

| System size | Starting dose            | Notes                                             |
|-------------|--------------------------|---------------------------------------------------|
| 5 gallons   | 1.0-1.5 ml, stir, retest | Ideal for bicarbonate pre-treatment of well water |

**WARNING**

**Concentrated acid in shared hardware.** Never mix concentrated phosphoric acid into a reservoir or line that has not been pre-flushed with clean water. Residual nutrient salts can react vigorously with concentrated acid, and a contaminated pump or emitter can fail catastrophically. Pre-flush, then dose into clean water, then add nutrients in the standard order.

**3.3 pH Up recipes**

Most Dragoon Mountain growers adjust down far more often than up — bicarbonate-rich well water and nitrifying-bacteria activity both push pH downward over time. But rainwater-fed systems and long vegetative runs can drift acidic, and you will eventually want a dependable pH-up option.

**3.3.1 Oak-ash lye (foraged, best local option)**

Oak woodland is the Dragoons' signature plant community, and oak ash — steeped in soft water — produces a potassium-rich alkaline liquid used historically for soap-making, tanning, and pickling. It is the most practical local pH-up source, and it has a useful side benefit: it adds potassium to your reservoir at the same time it raises pH.

**What you need:** fully-burnt oak ash from seasoned hardwood (no charcoal chunks, no resinous wood) · a non-metallic container · rainwater or RO water · fine cloth · a pH meter · gloves.

| Step | Action                                                             | Notes                                                    |
|------|--------------------------------------------------------------------|----------------------------------------------------------|
| 1    | Pack a non-metallic container loosely with cold oak ash            | Food-grade bucket or large glass jar                     |
| 2    | Pour 2-3 volumes of rainwater (or RO water) slowly through the ash | Drip through over several hours into a collection vessel |
| 3    | Let the lye settle overnight                                       | Dark amber-to-brown liquid; expect pH 11-13              |

| Step | Action                                                      | Notes                                                     |
|------|-------------------------------------------------------------|-----------------------------------------------------------|
| 4    | Strain through fine cloth into a labelled dark glass bottle | Discard the ash (spread on compost, not in hydro)         |
| 5    | Seal tightly                                                | Keeps indefinitely. Label: pH Up — Oak Ash Lye — CAUSTIC. |

**WARNING**

**Oak-ash lye is caustic.** Wear gloves and eye protection. Never splash on skin — it produces deep chemical burns. Dilute heavily before reservoir use; this is the strongest pH-up option in the guide. If splashed, rinse with running water for 15 minutes and seek medical attention if skin damage is present.

| System size | Starting dose                             | Notes                                                    |
|-------------|-------------------------------------------|----------------------------------------------------------|
| 1 gallon    | 0.1-0.3 ml lye, stir, wait 10 min, retest | Dilute lye 1:10 with water first for easier control      |
| 3 gallons   | 0.3-0.8 ml lye, stir, wait 10 min, retest | Drop by drop with a pipette                              |
| 5 gallons   | 0.5-1.5 ml lye, stir, wait 10 min, retest | Pre-dilute to a 1:10 working solution and dose from that |

**LOCAL TIP**

Oak-ash lye adds potassium (K) — a macronutrient you want. This makes it the least disruptive pH-up choice for a nutrient program: every correction doubles as a small top-up of K. The one crop where this matters most is fruiting peppers and tomatoes, which benefit from extra potassium in the flowering and sizing windows anyway. A winter's *Q. emoryi* fireplace ash will supply pH-up for a full season.

**3.3.2 Hydrated lime from caliche or limestone (foraged, backup)**

Caliche (calcium carbonate hardpan) and limestone outcrops throughout Cochise County can be processed into hydrated lime — calcium hydroxide — which is a strong pH-up material that also contributes calcium. It is more work than oak-ash lye and less kind to the skin, but it is a useful backup.



| Step | Action                                                                      | Notes                                                    |
|------|-----------------------------------------------------------------------------|----------------------------------------------------------|
| 1    | Burn limestone or caliche in a hot fire (bonfire or kiln) for several hours | Drives off CO <sub>2</sub> ; leaves quicklime (CaO)      |
| 2    | Slake the quicklime by slowly adding a small amount of water                | It will hiss and steam                                   |
| 3    | Keep adding water until it forms a smooth paste                             | Calcium hydroxide (hydrated lime)                        |
| 4    | Stir a teaspoon of paste into 1 cup of water. Let settle                    | Clear liquid above the sediment is “lime water” — pH 12+ |
| 5    | Dose the clear supernatant drop by drop                                     | Very strong                                              |

**WARNING**

**Quicklime reacts violently with water and can burn skin severely.** Protective gear is non-optional. If you can avoid this method with oak-ash lye, do.

**3.3.3 Potassium hydroxide stock (purchased, closest to commercial)**

Commercial pH Up is typically a dilute potassium-hydroxide solution. Food-grade potassium hydroxide (KOH) pellets — the same material used in soap-making — are widely available from soap-supply shops and some homebrew suppliers. As with phosphoric acid, you can make the exact commercial product at a fraction of the cost.

| Step | Action                                                                        | Notes                                                                                   |
|------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1    | Purchase food-grade KOH pellets                                               | Often labelled “potassium hydroxide food grade” or “caustic potash”. Store dry, sealed. |
| 2    | Gloves, eye protection, ventilation                                           | KOH pellets are strongly caustic and hygroscopic                                        |
| 3    | Slowly add 1 tbsp (≈10 g) of KOH pellets to 1 cup (240 ml) of distilled water | Solution will warm noticeably. Stir with glass or plastic (not metal).                  |
| 4    | Bottle and label                                                              | Dark glass or thick HDPE. Label: pH Up — KOH 4 % — CAUSTIC.                             |

| Step | Action                    | Notes                                    |
|------|---------------------------|------------------------------------------|
| 5    | Dose 0.3–0.5 ml at a time | Stir thoroughly, wait 10 minutes, retest |

  

| System size | Starting dose            | Notes                                    |
|-------------|--------------------------|------------------------------------------|
| 1 gallon    | 0.1–0.3 ml, stir, retest | Adds useful potassium, like oak-ash lye  |
| 3 gallons   | 0.3–0.8 ml, stir, retest | Drop by drop                             |
| 5 gallons   | 0.5–1.5 ml, stir, retest | Clean, consistent, commercial-equivalent |

**WARNING**

**KOH pellets are hygroscopic and exothermic in water.** Add pellets to water (never the reverse), in a heat-resistant glass or HDPE container, slowly, with stirring. The solution will warm to 130–150 °F during dissolution — let it cool before bottling.

**3.3.4 Baking-soda solution (purchased, emergency only)**

Sodium bicarbonate (baking soda) raises pH gently and is safe to handle, but it adds sodium — which competes with potassium and calcium uptake and accumulates in recirculating systems. It is fine as an occasional emergency fix, but **not as a regular pH-up strategy**.

| System size | Starting dose (½ tsp baking soda in 1 cup water, then dose the liquid) | Notes                                        |
|-------------|------------------------------------------------------------------------|----------------------------------------------|
| 1 gallon    | 2–5 ml of solution, stir, retest                                       | Change reservoir within the week to flush Na |
| 3 gallons   | 5–15 ml, stir, retest                                                  | Not for long runs                            |
| 5 gallons   | 10–25 ml, stir, retest                                                 | Consider oak-ash lye or KOH stock instead    |

**3.4 Comparison table****pH Down options compared**

| Option                    | Strength (pH)  | Cost                       | Shelf life          | Best used for                                            | Rating |
|---------------------------|----------------|----------------------------|---------------------|----------------------------------------------------------|--------|
| Prickly-pear vinegar      | Mild 3.0–3.5   | Free                       | 3–6 mo refrigerated | Daily micro-adjustments; minor rainwater tweaks          | ●●●●○  |
| Fermented citric acid     | Medium 2.0–3.0 | Free                       | 1–2 wks             | Moderate corrections; better than vinegar for well water | ●●●○○  |
| Sulfur-dioxide acid water | Strong 1.5–3.0 | Free                       | 24 hrs only         | Bicarbonate pre-treatment of large well-water volumes    | ●●●○○  |
| Citric-acid stock         | Medium 2.0–2.5 | ~\$5/lb (lasts 1–2 yrs)    | 12+ mo sealed       | <b>Everything — best all-round option</b>                | ●●●●●  |
| White vinegar             | Mild 2.5–3.0   | ~\$2/bottle                | 2+ yrs sealed       | Quick adjustments; fine for rainwater systems            | ●●●●○  |
| Phosphoric acid (diluted) | Strong 1.0–2.0 | ~\$15/bottle (lasts years) | 2+ yrs sealed       | Well-water pre-treatment; closest to commercial pH Down  | ●●●●●  |

### pH Up options compared

| Option                  | Strength (pH) | Cost                   | Shelf life        | Best used for                                     | Rating |
|-------------------------|---------------|------------------------|-------------------|---------------------------------------------------|--------|
| Oak-ash lye             | Strong 11-13  | Free                   | Indefinite sealed | <b>The Dragoons default pH-up. Adds useful K.</b> | ●●●●●  |
| Hydrated lime (caliche) | Strong 12+    | Free                   | Indefinite dry    | Ca-plus-pH-up when deficient in both              | ●●●○○  |
| KOH stock               | Strong 13+    | ~\$10/lb (lasts years) | 12+ mo sealed     | Commercial-equivalent precision. Adds K.          | ●●●●●  |
| Baking soda             | Mild 8-9      | ~\$1/box               | 2+ yrs sealed     | Occasional emergency only — adds Na               | ●●○○○  |

### 3.5 Universal dosing protocol

This sequence applies to any acid or base in §3.2 or §3.3.

| Step | Action                                 | Notes                                                                          |
|------|----------------------------------------|--------------------------------------------------------------------------------|
| 1    | Fill reservoir with source water       | Note starting pH and EC                                                        |
| 2    | Bicarbonate check                      | If well water, add acid slowly until fizzing stops completely. See Guide 3 §4. |
| 3    | Add nutrients in order                 | <i>Masterblend</i> → Epsom → Cal-Nitrate (or Part 2 equivalents)               |
| 4    | Measure pH                             | Likely 6.5–7.5 after nutrients                                                 |
| 5    | Dose chosen pH adjuster 1 ml at a time | Stir thoroughly after each addition                                            |
| 6    | Wait 5-10 minutes between additions    | pH continues to shift after mixing                                             |
| 7    | Target pH 5.8-6.2                      | Sweet spot for most crops                                                      |

| Step | Action                    | Notes                 |
|------|---------------------------|-----------------------|
| 8    | Daily check every morning | Small doses as needed |

**LOCAL TIP**

**Recommended Dragoons hobby-grower setup.** The pragmatic kit for most growers in Cochise County:

- **Citric-acid stock** (3.2.4) — precise daily pH-down adjustments and fine-tuning after nutrients are added.
- **Prickly-pear vinegar** (3.2.1) — minor corrections, made fresh from the landscape each September.
- **Oak-ash lye** (3.3.1) — your pH-up, made fresh each year from last winter's fireplace.
- **Phosphoric acid** (3.2.6) — for the initial bicarbonate pre-treatment if your well water is very hard (pH 8+).

Keep two bottles of each on hand — one on the bench, one as the backup — because the one moment you absolutely need a working pH-down dropper is the moment the bottle is empty.

## Part 4 — Indoor environment & precision

### LED · VPD · CO<sub>2</sub> · sensors · automation

When the system moves indoors or under a roof — a garage, a basement, a closed greenhouse, a converted spare room — nutrition alone no longer sets the ceiling on yields. **Light, vapour pressure deficit (VPD)**, and **CO<sub>2</sub>** become the dominant variables. This part covers the precision side of indoor growing in the Dragoons, where 300+ sunny days and large diurnal swings make supplemental lighting and environmental control unusually high-leverage.

### 4.1 LED basics

Modern quantum-board LEDs have transformed indoor growing. They deliver full-spectrum light at high efficiency (2.5–3.0  $\mu\text{mol/J}$ ), produce far less heat than HPS, and can be tuned to the exact spectrum each crop needs. In the Dragoons, LEDs also allow year-round growing independent of season or weather.

#### Key light metrics

| Term           | What it means                                                       | Why it matters                                                                                   |
|----------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| PPFD           | Photosynthetic Photon Flux Density ( $\mu\text{mol/m}^2/\text{s}$ ) | Instantaneous light intensity at canopy. The most important number.                              |
| DLI            | Daily Light Integral ( $\text{mol/m}^2/\text{day}$ )                | Total light per day. $\text{PPFD} \times \text{hours} \div 3600 \times 0.0036$ . Varies by crop. |
| PPE / Efficacy | $\mu\text{mol}$ of light per joule of electricity                   | Higher = more efficient. Quality LEDs: 2.5–3.0 $\mu\text{mol/J}$ . HPS: $\sim 1.7$ .             |
| Spectrum       | Wavelength blend (nm)                                               | Full spectrum (3000 K + 5000 K) for veg; add red (660 nm) for fruiting.                          |
| Photoperiod    | Hours of light per 24-hr cycle                                      | Veg: 18 on / 6 off. Fruiting: 12–16 hrs.                                                         |

#### PPFD targets by crop and stage

| Crop / stage                 | PPFD ( $\mu\text{mol/m}^2/\text{s}$ ) | DLI ( $\text{mol/m}^2/\text{day}$ ) | Photoperiod |
|------------------------------|---------------------------------------|-------------------------------------|-------------|
| Seedlings / propagation      | 100–200                               | 5–10                                | 18 hrs      |
| Leafy greens (lettuce, kale) | 200–400                               | 12–17                               | 16–18 hrs   |
| Herbs (basil, cilantro)      | 300–500                               | 15–20                               | 16–18 hrs   |

| Crop / stage        | PPFD ( $\mu\text{mol}/\text{m}^2/\text{s}$ ) | DLI ( $\text{mol}/\text{m}^2/\text{day}$ ) | Photoperiod |
|---------------------|----------------------------------------------|--------------------------------------------|-------------|
| Tomato — vegetative | 400-600                                      | 20-25                                      | 18 hrs      |
| Tomato — fruiting   | 600-900                                      | 25-35                                      | 14-16 hrs   |
| Pepper — vegetative | 400-600                                      | 18-25                                      | 18 hrs      |
| Pepper — fruiting   | 600-800                                      | 22-30                                      | 14-16 hrs   |
| Cucumber — fruiting | 700-1000                                     | 30-40                                      | 14-16 hrs   |

### Recommended LED fixtures (2025)

| Light                 | Power | Coverage | Efficacy                      | Best for                            |
|-----------------------|-------|----------|-------------------------------|-------------------------------------|
| Spider Farmer SF-4000 | 400 W | 4×4 ft   | 2.9 $\mu\text{mol}/\text{J}$  | Tomatoes, peppers, high-light crops |
| Mars Hydro FC-E4800   | 480 W | 4×4 ft   | 2.85 $\mu\text{mol}/\text{J}$ | Heavy fruiting crops                |
| Gavita Pro 1700e LED  | 645 W | 5×5 ft   | 2.6 $\mu\text{mol}/\text{J}$  | Commercial-scale fruiting           |
| Spider Farmer SE-3000 | 300 W | 3×4 ft   | 2.7 $\mu\text{mol}/\text{J}$  | Mixed veg + herb production         |
| HLG 100 V2 (QB)       | 100 W | 2×2 ft   | 2.7 $\mu\text{mol}/\text{J}$  | Seedlings, small herb setups        |

### Dragoon advantage — supplemental lighting

At 4,500–7,500 ft elevation, natural sunlight intensity is 15–20 % higher than at sea level. For greenhouse or semi-outdoor growing, this means natural DLI during fall and spring can reach 40–50  $\text{mol}/\text{m}^2/\text{day}$  — more than enough for any crop. LEDs are most valuable in winter (low sun angle) and for extending photoperiod after sunset. Just 2–4 hours per day of supplemental LED lighting in a greenhouse dramatically boosts winter yields.

#### LOCAL TIP

Use shade cloth and LEDs together: shade cloth for midday UV protection during the May–June peak, LEDs for early-morning and evening photoperiod extension. At 4,700 ft on the east side of the Dragoons, this combination gives near-equatorial DLI through eight months of the year without any cooling or supplemental heating.





Figure 1: An indoor kitchen-herb bench under a dedicated LED panel. Mint, rosemary, thyme, dill, and chives all run on the same fixture and the same 16-hr photoperiod; the WiFi controller on the wall handles the timer plus humidity alerts referenced in §4.4.

## 4.2 VPD and CO<sub>2</sub> management

### VPD — Vapour Pressure Deficit

VPD measures the difference between how much moisture the air holds and how much it could hold. Too low = sluggish transpiration, wet leaves, fungal risk. Too high = plants close stomata, nutrient uptake stops, wilting. VPD is the single most overlooked variable in small-scale growing — getting it right unlocks faster growth, better nutrient uptake, and dramatically healthier plants.

| Growth stage             | Temperature         | Humidity | Target VPD (kPa) |
|--------------------------|---------------------|----------|------------------|
| Seedlings / clones       | 72-77 °F (22-25 °C) | 70-80 %  | 0.4-0.8          |
| Early vegetative         | 72-79 °F (22-26 °C) | 60-70 %  | 0.8-1.0          |
| Late vegetative          | 75-82 °F (24-28 °C) | 55-65 %  | 1.0-1.2          |
| Flowering / fruiting     | 68-79 °F (20-26 °C) | 45-55 %  | 1.2-1.6          |
| Late fruiting / ripening | 65-75 °F (18-24 °C) | 40-50 %  | 1.4-1.8          |



### Dragoon climate interaction

During the monsoon (July–September), outdoor humidity runs 50–70 %, so VPD drops naturally. Reduce watering frequency and watch for *Botrytis* and powdery mildew. In the dry season (October–June), humidity often sits at 10–30 % and VPD is very high — add a humidifier to your grow space or plants will constantly stress. At elevation, UV is 15–20 % stronger than at sea level, which accelerates leaf-surface drying in a greenhouse with natural light.

### Temperature & CO<sub>2</sub>

| Parameter                  | Optimal range            | Dragoon note                                                                                              |
|----------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Air temp (lights on)       | 72–82 °F (22–28 °C)      | Summer nights naturally cool — great for fruiting crops that want cold nights                             |
| Air temp (lights off)      | 62–70 °F (17–21 °C)      | Dragoon nights often ideal — use outdoor air exchange                                                     |
| Root zone temp             | 65–72 °F (18–22 °C)      | <b>Most critical parameter.</b> Bury or insulate reservoirs. Higher = root rot.                           |
| CO <sub>2</sub> (ambient)  | 400 ppm (natural)        | At elevation, ambient CO <sub>2</sub> is slightly lower — sealed rooms benefit from enrichment            |
| CO <sub>2</sub> (enriched) | 800–1200 ppm             | Boosts photosynthesis 20–30 %. Only effective with high PPFD (600+ $\mu\text{mol}/\text{m}^2/\text{s}$ ). |
| Airflow                    | Gentle constant movement | Strengthens stems, prevents hot spots, reduces fungal risk during monsoon                                 |

**CO<sub>2</sub> enrichment is only worth it if PPFD is already high.** At 400  $\mu\text{mol}/\text{m}^2/\text{s}$ , the plant cannot use extra CO<sub>2</sub> — light is the limiting factor. Push PPFD to 600+  $\mu\text{mol}/\text{m}^2/\text{s}$  *first*, then add CO<sub>2</sub>. In a sealed Dragoons grow room with a strong LED, 1000 ppm CO<sub>2</sub> is the canonical target.

### 4.3 Sensor priority list

What to buy first, in priority order. Cost ranges are 2025 USD.

| Device                                     | Function                                | Cost range | Priority                                 |
|--------------------------------------------|-----------------------------------------|------------|------------------------------------------|
| Digital pH/EC meter                        | Manual daily monitoring                 | \$30-150   | ●●●●● Essential                          |
| Digital timer (light cycle)                | Automates photoperiod                   | \$15-30    | ●●●●● Essential                          |
| Inkbird temp controller                    | Triggers fans/heaters at set temps      | \$25-50    | ●●●●● Essential                          |
| Digital hygrometer                         | Monitors temp + humidity for VPD        | \$15-25    | ●●●●● Important                          |
| WiFi grow controller (Inkbird, Trolmaster) | Remote monitoring, alerts to phone      | \$100-400  | ●●●● Remote-location value               |
| Auto top-off system                        | Maintains reservoir level               | \$50-150   | ●●●● Saves daily labour                  |
| Autopilot CO <sub>2</sub> controller       | Maintains CO <sub>2</sub> at target ppm | \$150-300  | ●●●● If using CO <sub>2</sub> enrichment |
| Dosatron / auto-doser                      | Automatically doses nutrients at set EC | \$150-500  | ●●●● Recommended at scale                |
| EC/pH dosing controller (Bluelab, Atlas)   | Continuously monitors and auto-doses    | \$500-1500 | ●●● Advanced / commercial                |

### Buy-it-first kit

For a new grower with a \$200 budget, this is the order:

1. **pH meter** (Apera PH60 or similar) — \$50
2. **EC meter** (Apera EC60 or HM Digital) — \$40
3. **Digital timer** (any reliable mechanical or smart timer) — \$20
4. **Hygrometer** — \$20
5. **Digital scale, 0.1 g** — \$15
6. Spare buffer solutions and calibration powder — \$30
7. Margin for replacements — \$25

This kit covers everything in Part 1's daily protocol. Automation comes after.

## 4.4 Automation

Automation turns a high-maintenance daily task into a manageable weekly check-in. For Dragoon Mountain growers — where temperature swings, monsoon humidity changes, and remote location can make daily attention difficult — even basic automa-

tion is a major quality-of-life improvement.

### Dragoon-specific automation priorities

- **Temperature controllers with phone alert** are high-value here. Large day/night swings mean reservoir temps can fluctuate significantly, and an Inkbird controller + WiFi plug + outdoor sensor catches the swing before it costs you a crop.
- **WiFi hygrometers with humidity alerts** are worth it during the monsoon. Rapid humidity swings July–September can push VPD out of range fast.
- **Off-grid alert systems** are valuable if you travel frequently. WiFi sensors that alert your phone when pH, temp, or EC goes out of range prevent crop losses on a remote property.
- **Light timers are non-negotiable.** Consistent photoperiod is critical for fruiting triggers and for preventing premature bolting in greens. Mechanical or smart, doesn't matter — just *consistent*.

### Solar-powered off-grid setup

#### LOCAL TIP

The Dragoons get 300+ sunny days per year. A modest solar + battery setup (400 W panels + 100 Ah lithium) can power LED lights, pumps, and all monitoring equipment — making this system effectively off-grid. A starter rig with two 200 W panels, a 100 Ah LiFePO4 battery, a 30 A charge controller, and a 1000 W inverter runs about \$800 and covers a 4-bucket bato setup with one 400 W LED indefinitely. Cross-reference Guide 3 §2 for the system build this powers.

### Automation troubleshooting

| Problem                             | Cause                                                 | Solution                                                                 |
|-------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
| Root rot (brown, slimy roots)       | Reservoir temp > 72 °F or insufficient O <sub>2</sub> | Insulate reservoir. More airstones. Beneficial bacteria (Hydroguard).    |
| White mineral crust on pots / media | Salt build-up from hard water                         | Flush with RO or rainwater. Switch primary source. Check EC baseline.    |
| pH unstable — rises rapidly         | Bicarbonates in source water                          | Pre-acidify before nutrients. Switch to RO or rainwater. See Guide 3 §4. |
| Slow growth despite good EC/pH      | VPD out of range or root zone too warm                | Check temp and humidity. Verify root zone < 72 °F.                       |

| Problem                          | Cause                                               | Solution                                                                 |
|----------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------|
| Powdery mildew on leaves         | High humidity + low airflow (common during monsoon) | Increase airflow. Reduce humidity below 60 %.<br>Remove affected leaves. |
| Light burn — bleached top leaves | LED too close or PPFD too high                      | Raise light height. Reduce intensity. Check manufacturer guide.          |
| Stretching / etiolation          | PPFD too low / not enough light                     | Lower LED. Increase intensity. Extend photoperiod to 18 hrs.             |
| Algae in reservoir               | Light getting into reservoir                        | Cover all surfaces. Opaque containers only.                              |

## Sources & references

---

*Reference material consulted, suggested for further reading, and where to verify or extend the claims in this guide.*

### Local resources

- **UA Cooperative Extension — Cochise County office (Sierra Vista)** — water-chemistry workups, bicarbonate testing, and the Tucson-area nutrient-management bulletins. Phone 520-458-8278.
- **Native Seeds/SEARCH (Tucson)** — heritage seed adapted to high-desert hydroponic systems; the upstream source for the pepper, basil, and tomato varieties this nutrient program is tuned to.
- **Sulphur Springs Valley feed and farm supply** — bulk *Masterblend 4-18-38*, calcium nitrate, and Epsom salt at agricultural rather than hobby pricing.
- **Local sources for Part 2 (DIY) materials** — Emory oak windfall on Dragoon foothill slopes (potash), Cochise County mine-adit roosts in winter only (guano), Sulphur Springs Valley caliche outcrops (calcium), Sulphur Springs Valley playas and seeps (*epsomite*), Bisbee and Tombstone mineral districts (copper, zinc trace).

### Books & references

- *Hydroponic Food Production* by Howard Resh — the standard commercial reference. The 116 / 47 / 188 ppm *Masterblend*-equivalent profile traces back to recipes worked out in this lineage.
- *Hydroponics: A Practical Guide for the Soilless Grower* by J. Benton Jones — the deepest book on EC, pH, and nutrient-uptake chemistry for small systems.
- *Teaming with Nutrients* by Jeff Lowenfels — the plant-physiology side of nutrient uptake (the *why* behind ratios), useful for Part 2 readers crossing over from soil growing.
- *Growing Food in a Hotter, Drier Land* by Gary Paul Nabhan — Sonoran-Desert context for Part 2 (local-sourcing) and the Cochise County mineral landscape.
- *The Whole-Brain Hydroponic Gardener* by various authors (eds.) — practical recipes for foraged and homemade pH adjusters and DIY micronutrient sources.

### Online resources

- **Cornell Controlled Environment Agriculture** ([cea.cals.cornell.edu](http://cea.cals.cornell.edu)) — peer-reviewed PPFD, DLI, and VPD targets by crop. The numerical targets in Part 4 §4.1 align with this dataset.
- **Bluelab knowledge base** ([bluelab.com](http://bluelab.com)) — practical EC/pH meter calibration, drift management, and reservoir-monitoring guides.
- **UA Cooperative Extension publications** ([extension.arizona.edu](http://extension.arizona.edu)) — SE Arizona water-chemistry, alkalinity, and salinity bulletins.
- **OMRI (Organic Materials Review Institute)** ([omri.org](http://omri.org)) — verify any input —

*Masterblend*, calcium nitrate, citric acid, hydrogen peroxide, KOH — for organic certification before applying.

- **DLI Calculator (Heliospectra, Fluence)** — convert PPFD × photoperiod into DLI for Part 4 lighting decisions.

## Closing

*One nutrient profile · four roads to get there*

This guide started as three documents — a Masterblend precision guide, a locally-sourced nutrients guide, and a pH up/down replacements guide — and consolidated them into a single four-part reference because the divisions between them were always artificial. The same grower who weighs *Masterblend* on Monday is foraging *epsomite* on Saturday. The same reservoir that takes purchased phosphoric acid in February takes oak-ash lye in November. The chemistry doesn't care which document the recipe lives in.

What this guide does add, beyond consolidation, is the resolution of one canonical conflict from the older library: **peppers at fruiting take 2.4 g/gal Epsom salt, not 1.5**. Some upstream Masterblend tables averaged peppers and tomatoes together and undercounted the magnesium that peppers actually use. Guide 2 (Peppers in the Dragoons) and Guide 3 (Bato Buckets) both reference the 2.4 g/gal figure; this guide is the home for the corrected recipe.

With *Masterblend's* precision, modern high-efficacy LED lighting, and controlled-environment techniques, the Dragoon Mountains become an exceptional growing location. Cooler summers than the low desert, abundant solar energy, monsoon rainwater, and long clear fall seasons all work in your favour. Keep the pepper Epsom at 2.4 g/gal once fruit is sizing, hold pH at 5.8-6.2, and the ceiling on your yields is set by VPD and light — not by nutrition.

### RELATED GUIDES

This guide is part of the Dragoon Mountains Growing Library: - **Guide 1 — Hydroponic Primer & Library Map** · front door, master Dragoons calendar - **Guide 2 — Peppers in the Dragoons** · soil + hydroponic + staking - **Guide 3 — Bato Bucket Hydroponic Systems** · system build, water treatment, daily operations - **Guide 4 — Hydroponic Nutrients** (this guide) - **Guide 5 — Grapes in the Dragoons** · Cochise County viticulture - **Guide 6 — Cool-Season Crops** · brassicas + alliums + greens + roots